

Isha Pawar

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Education

Purdue University

Jan 2025 - May 2026

Masters of Science in Computer Science

CGPA: 4/4

Relevant Coursework: Analysis of Algorithms, NLP, Deep Learning, Application Development, Database

Sinhgad College of Engineering

June 2018 - May 2022

Bachelors of Engineering in Computer Science

CGPA: 3.9/4

Relevant Coursework: Operating Systems, Data structures, System Design, Object-oriented programming

Skills

Languages: Java, Python, JavaScript, TypeScript, C/C++

Frontend: React.js, Next.js, HTML5, CSS3, jQuery, AngularJS

Backend & Systems: Java, Spring Boot, REST APIs, Microservices, Object-Oriented Design, Distributed Systems, Event-Driven Architecture, Kafka, RabbitMQ

Cloud & DevOps: AWS, EC2, Lambda, S3, DynamoDB, Docker, Kubernetes, GitHub Actions, CI/CD, Elasticsearch

Databases: MySQL, PostgreSQL, MongoDB, DynamoDB, Oracle 11g

AI & ML: LLMs, Prompt Engineering, OpenAI API, LangChain, RAG, Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy

Work Experience

Research Assistant, Purdue University

Oct 2025 - May 2026

- Designed prompt-based experiments to evaluate consistency, multimodal alignment, and semantic fidelity across Vision-Language Models using varied prompt formulations.
- Built an image-text-image evaluation workflow to test semantic accuracy, reconstruction quality, and model behavior across multimodal AI outputs.

Software Engineer, Standard Chartered Bank

Sep 2022 - Dec 2024

- Designed and deployed high-availability backend services using Java, Spring Boot, Docker, and Kubernetes, improving payment-processing efficiency across distributed payment systems.
- Automated release validation and trend reporting in Alteryx, reducing downtime by 25% and release issues by 15%.
- Implemented event-driven workflows using AWS Lambda, DynamoDB, Kafka, and RabbitMQ for low-latency, asynchronous backend processing.
- Built CI/CD pipelines for automated deployment to AWS EC2, improving release reliability, rollback readiness, and backend iteration speed.
- Optimized AWS costs by \$37K by identifying unused AWS resources and establishing S3 bucket expiration policies.
- Developed Python scripts to monitor deployment logs, detect release anomalies, and improve operational visibility for backend services.
- Participated in backend design reviews and code reviews, improving service reliability, maintainability, and adherence to engineering standards.

Software Engineer Intern, Persistent Systems

Jan 2022 - Jul 2022

- Diagnosed and resolved software issues through root cause analysis, improving efficiency and system reliability.
- Optimized SQL-based school datasets using indexing and data-cleaning techniques, reducing redundancy by 17%.
- Enhanced 12 React components by integrating Redux, Material-UI, and Formik to improve usability and maintainability.

Project Work

AI-Powered Video Transcriber

Aug 2025 - Dec 2025

- Built an AI-powered video transcriber using Faster-Whisper ASR and a custom T5 NLP model (100K+ sentence pairs)
- Designed a low-latency AI data pipeline using FFmpeg, Faster-Whisper ASR, NLP cleanup, and subtitle generation, optimizing for accuracy, privacy, and local inference performance.
- Developed local, cross-platform desktop application (Windows/macOS) preserving privacy and cloud-free inference.

MastodonGPT

Jan 2025 - May 2025

- Built an interactive React/Next.js chatbot using Gemini 2.0, LangChain, and RAG to help users navigate university resources through retrieval-backed responses.
- Served 9,824 users after launch and improved chatbot responses based on feedback from student usability testing.
- Integrated FAISS vector database with SQL storage for accurate, low-latency responses and developed the full-stack solution using React.js, Next.js, and Python.

Certifications

AWS Certified Cloud Practitioner, AWS Certified Solutions Architect Associate, Microsoft Azure Fundamentals, Microsoft Python Programming